

Section 05



Industrial Security (Bachelor)

Section 05 – Threat Landscape

Threat Landscape

Lecture Notes

Department of Computer Science

1 Threat Actors and Frameworks

2 Landmark Incidents

3 Cross-Cutting Risks

By the end of this section you will be able to

- Categorise threat actors targeting industrial environments.
- Map ICS attacks to the MITRE ATT&CK for ICS matrix.
- Analyse landmark incidents: Stuxnet, Ukraine, TRITON, Colonial.
- Identify supply-chain and insider risk vectors.

1

Threat Actors and Frameworks

Section 05 – Threat Landscape

Nation-state APTs – Sandworm, Volt Typhoon, Lazarus

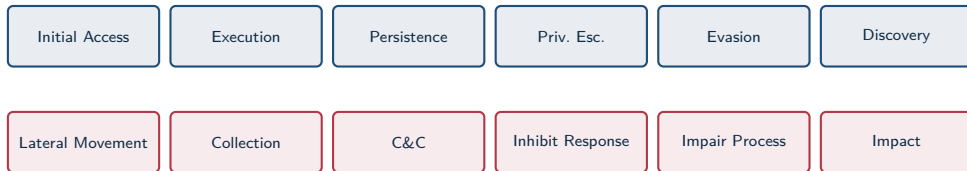
Cybercrime / Ransomware – LockBit, BlackCat, Conti

Hacktivists – CyberAv3ngers, GhostSec

Insiders – disgruntled employees, contractors

Opportunists – script kiddies, Shodan hunters

Competitors – industrial espionage



Why this matters

ATT&CK for ICS gives a shared vocabulary for techniques, detections, and incident discussion across teams and vendors.

2

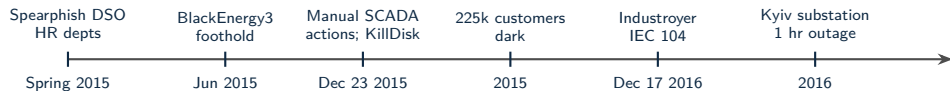
Landmark Incidents

Section 05 – Threat Landscape

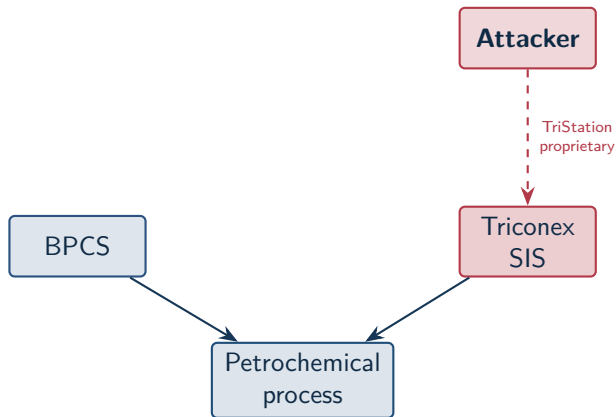


Natanz, 2010

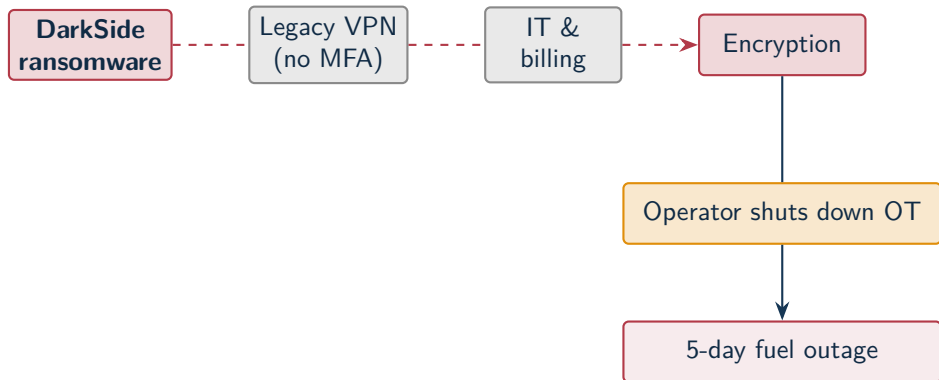
Four zero-days, two stolen code-signing certificates. Subtly destroyed ~1 000 centrifuges over months while feeding the HMI *normal-looking* data. The first widely-known cyber-physical weapon.



Sandworm (GRU). Industroyer was the first malware framework with built-in grid protocol modules.



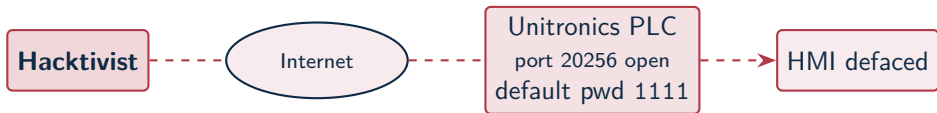
First publicly known attack on a Safety Instrumented System. Programming error in the malware tripped the plant – and exposed the operation. Watershed moment: attackers reached the safety layer.



IT compromise; operator pre-emptively halted OT. Highlights IT/OT interdependence and the physical impact of ransomware.

Lessons that keep coming back

Default credentials and **direct Internet exposure** of PLCs are still common – and still get exploited by low-skill actors at scale.



No process damage, but trivially exploited at multiple water utilities. **Reminder:** never expose OT to the Internet, always change defaults.

3

Cross-Cutting Risks

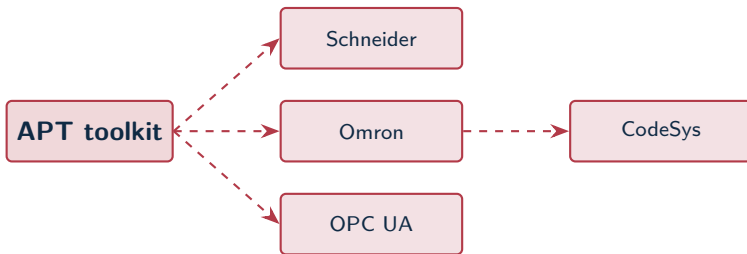
Section 05 – Threat Landscape



SolarWinds, 3CX, MOVEit – and OT-relevant cases such as Codesys runtime CVEs (*CoDe16*).



Maroochy Shire (2000): ex-contractor with a stolen radio modem released sewage 46 times.



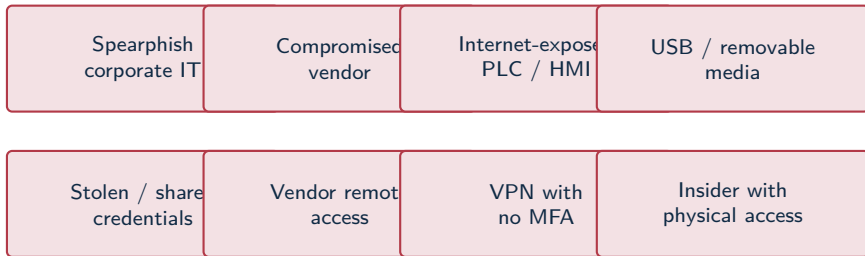
First multi-target ICS framework

PIPEDREAM is the first modular framework with pre-built modules for several vendors at once – captured before it was used in anger.

- Suspected China-linked APT, active since at least 2021.
- Targets US critical infrastructure: telecoms, water, energy.
- Uses *built-in tools* (PowerShell, WMI, native Windows) – minimal new malware.
- Apparent goal: pre-positioning for disruptive action during conflict.

Caution

The hardest threat to detect is the one that looks like a normal admin task. Behavioural analytics matter more than signatures.



★ Key Takeaway: What to remember from this section

- Threat actors range from script kiddies to nation-state APTs, each with different capabilities.
- MITRE ATT&CK for ICS provides a shared vocabulary for techniques and detections.
- Stuxnet → Ukraine → TRITON shows an escalating willingness to harm.
- Supply chain and insider risk demand controls beyond the perimeter.

End of Section 05



Threat Landscape

Questions and discussion